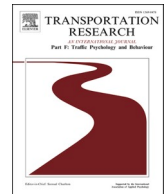




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Estimation of driver vigilance level for various cognitive distractions when drivers use advanced driving assistance functions

Mengjiao Wu^{a,b}, Xuesong Wang^{a,b,c,*}, Chris Lee^d, Shikun Liu^{a,b}, Jiawen Chen^{a,b}, Yiran Sun^e

^a College of Transportation, Tongji University, Shanghai 201804, China

^b The Key Laboratory of Road and Traffic Engineering, Ministry of Education, Shanghai 201804, China

^c Shanghai Pudong New Area Mental Health Center, School of Medicine, Tongji University, China

^d Department of Civil & Environmental Engineering, University of Windsor, 401 Sunset Ave, Windsor, Ontario N9B 3P4, Canada

^e Alliance Manchester Business School, University of Manchester, United Kingdom

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ABSTRACT

According to the World Health Organization report, 162 countries have enacted legislation prohibiting distracted driving. The regulations on distracted driving mainly restrict physical and visual distractions related to the use of handheld phones, with less restrictions on cognitive distractions. However, it is not yet clear which cognitive distracted driving behaviors may lead to significant decrease in driver vigilance. The purpose of this study is to assess how different cognitive distractions affect driver vigilance when drivers use advanced driving assistance functions in a monotonous driving highway environment. Using a driving simulator and external devices, the multi-dimensional including mental workload, vigilance reaction time and accuracy, eye movement data were collected from 31 participants. Cognitive distractions types included visual-manual-cognitive distraction tasks, auditory-cognitive distraction tasks and a no-distraction task. Three Multiple Criteria Decision-Making methods were used to evaluate the driver vigilance level during different types of cognitive distractions. The results indicate that participants exhibited lower vigilance levels during visual-manual-cognitive distraction tasks (L3 and L2) compared to auditory-cognitive distraction tasks (L2 and L1) and the no-distraction task (L1). This was due to the nature of the visual-manual-cognitive tasks, such as the message-sending task, which involve both language generation and comprehension, requiring higher levels of brain activation and attention resources. Therefore, visual-manual-cognitive distraction tasks should be limited when drivers use advanced driving assistance functions. In contrast, simple auditory-cognitive distractions, such as listening to new tasks, helped drivers remain alert and maintained vigilance levels comparable to the no-distraction task. This indicates that the auditory-cognitive distractions that involve language comprehension can assist drivers in maintaining an appropriate stimulation level, thereby preventing a decrease in vigilance associated with mind-wandering. Therefore, auditory-cognitive distractions involving language comprehension can be permitted in simple driving environments. As the vigilance level was lower at moderate level for auditory-cognitive involving working memory(1-back), it is recommended to warn this type of distraction.

* Corresponding author.

E-mail address: wangxs@tongji.edu.cn (X. Wang).

1. Introduction

Distracted driving is one of the important factors affecting road safety (National Center for Statistics and Analysis, 2020). In particular, during autonomous driving, drivers are more susceptible to distraction, and their vigilance to hazardous road stimuli tends to diminish over time (Greenlee et al., 2018). As a result, they may be unable to provide reliable supervision of the vehicle's interaction with the road environment. To address this issue, the Society of Automotive Engineers (2021), the European Union (2019), and the Chinese government (State Administration for Market Regulation, 2022) have each issued standards and reports emphasizing the importance of monitoring drivers while advanced driving assistance systems are in use. These guidelines aim to assess driver vigilance, ensure the driver's capacity to drive, and enhance road safety. Distracted driving is the key monitoring content of the EU driver status system.

Although the World Health Organization report indicates that 162 countries have enacted legislation prohibiting distracted driving (World Health Organization, 2023), the regulations primarily address physical and visual distractions related to the use of handheld phones, with less focus on cognitive distractions. In fact, it is still unclear which types of cognitive distraction significantly reduce driver vigilance level when advanced driving assistance system functions are used. It is also not clear which type of cognitive distracted driving may help keep drivers at high driver vigilance level in monotonous driving environments. According to resource control theory (Fortenbaugh et al., 2017), in a monotonous driving environment, the executive control exerted by the driver gradually weakens as driving time increases. The theory suggests that internal mind-wandering may represent an individual's automated processing state.

Attention resources are constantly competed for by both mind-wandering and driving tasks, with an increasing proportion being allocated to mind-wandering, which in turn results in a decrease in driving vigilance.

Previous studies have predominantly assessed the risk of distracted driving based on a single type of indicator. For example, an improved criterion-based method was employed to analyze the risk level of driving distractions associated with phone use (including voice and video calls) using four eye movement indicators (Wang et al., 2023). However, risk assessments based on a single type of indicator reflect only one aspect of the driver's vigilance state (such as visual processing) and fail to provide a comprehensive view of overall driver vigilance. In this context, the Multiple Criteria Decision Making (MCDM) method can be applied to evaluate vigilance levels more comprehensively, taking into account multiple types of indicators and the allocation of driver attention.

Thus, the objective of this study is to evaluate the decrease in driver vigilance caused by different types of cognitive distraction using the MCDM method. To achieve this, multidimensional data were collected from a series of driving simulator experiments. Three MCDM methods – the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), Fuzzy Comprehensive Evaluation, and Rank-Sum Ratio (RSR) – were applied to evaluate the comprehensive vigilance scores and vigilance levels. This study aims to provide a theoretical basis for limiting distracted driving behaviors in regulations and standards implemented in driver monitoring systems, such as the development of multi-level warnings to prevent the abuse or misuse of advanced driving assistance functions.

2. Literature review

2.1. Driver vigilance and distracted driving

There is currently no widely accepted definition of vigilance. Previous studies on vigilance have primarily focused on drivers' perception of stimuli and the retention of attention. Mackworth described vigilance as a psychological readiness to perceive and respond (Mackworth, 1948). Davies & Parasuraman, (1982) defined vigilance as a cognitive process related to an individual's ability to maintain sustained attention while monitoring for signals over an extended period. Based on research utilizing brain imaging techniques, attention has been classified into selective attention, executive attention, and attentional vigilance, the latter of which refers to the process by which individuals maintain concentration on a specific stimulus (Posner & Rothbart, 2007).

Distracted driving is a major cause of reduced driver vigilance. When drivers are distracted, their attention shifts to activities unrelated to driving, leading to a decrease in their ability to process driving-related information. The driver distraction guidelines of the U. S. National Highway Traffic Safety Administration (NHTSA) classify distractions into four categories: visual, manual, cognitive, and auditory (National Highway Traffic Safety Administration, 2012). Visual distraction occurs when drivers look away from the road to gather visual information. Manual distraction occurs when drivers take a hand off the steering wheel. Cognitive distraction occurs when drivers divert their mental attention from the driving task. Auditory distraction occurs when drivers pay attention to auditory signals and ignore the road environment.

In NHTSA's cognitive load-related naturalistic driving research, cognitive distraction is categorized into language-generation tasks, language-comprehension tasks, memory tasks and other tasks that impose cognitive load (Angell et al., 2015).

Language-generation tasks involve the creation of content, its organization, expression through words, and sequential speech production. Language-comprehension tasks encompass the reception of auditory input, parsing, and understanding, which may also engage long-term and working memory. Memory tasks involve processes such as working memory, retrieval, and occasionally the engagement of executive attention. Other tasks that impose cognitive load include visual-manual tasks.

2.2. Indicators of driving vigilance

Various types of indicators can be used to assess driver vigilance, including not only eye movement indicators but also reaction time, mental workload, and pulse rate.

As confirmed by Trichle's study, 83 % of the environmental information received by drivers is dependent on visual perception (Trichler, 1967). Therefore, eye movement is a crucial indicator of driver vigilance, reflecting the efficiency with which visual information is processed. The duration of glances on the road measures the processing of road information from both a temporal and spatial perspective, and this indicator shows performance differences among various types of vigilance decrease. Driving vigilance decreases when drivers engage in visual-manual tasks, leading to a reduction in the duration of glances on the road (Wang et al., 2017). In contrast, auditory-cognitive tasks are associated with longer glances on the road, although they also decrease driving vigilance (Angell et al., 2015). Additionally, fatigue-induced decreases in vigilance are linked to shorter glance durations, accompanied by an increase in PERCLOS (Wang & Xu, 2016).

The visual span on the road measures the processing of road information from a spatial perspective, and this indicator also varies in performance across different types of vigilance decrease. A decrease in the visual span on the road has been observed during cognitive tasks when compared to baseline levels (Kountouriotis et al., 2016). Visual span can also be analyzed from both horizontal and vertical perspectives (Robbins & Chapman, 2019). A smaller horizontal and vertical spread of search was observed during cognitive, visual, and visual-cognitive tasks compared to baseline (Liang & Lee, 2010). As the difficulty of a secondary (non-driving related) task increased, the horizontal spread of search increased for visual distraction tasks, while it decreases for cognitive distraction tasks (Son & Park, 2021).

Reaction time is a critical indicator of driving vigilance. Previous studies have consistently shown that a decrease in driving vigilance is typically associated with an increase in reaction time, which can be attributed to various factors, including fatigue (Zhang, 2022), distraction (Karthauss et al., 2020), and prolonged monitoring (Greenlee et al., 2018). The Detection Response Task (DRT) has been widely used to measure vigilance response time and hit rate. DRT is also an effective method for assessing the attentional effects of cognitive load imposed on an individual by secondary tasks (Kristina & Jaka, 2018). In accordance with ISO guidelines, sensory stimulation is provided to the driver every 3–5 s, and the driver is required to respond by pressing a button. The duration of sensory stimulus presentation is 1 s. For a response to be considered valid, it must be initiated within 100–2,500 ms after the stimulus is presented (International Organization for Standardization, 2022). The DRT is simple, does not impose additional cognitive load, and is suitable for controlled driving environments, such as driving simulators with variable driving settings. In this study, the DRT was used as a tool to measure vigilance status, reflecting the efficiency of drivers' response to warning stimuli during distracted tasks.

Mental workload refers to the balance between a driver's limited cognitive resources and the demands of the task at hand (Virtanen et al., 2022). When mental workload exceeds the driver's cognitive capacity, it can lead to decreased task performance (Wickens, 1984). Therefore, mental workload indicators can be used to assess a driver's vigilance status. National Aeronautics and Space Administration Task Load Index (NASA-TLX) is a widely used subjective method for evaluating mental workload, encompassing six dimensions: mental demand, physical demand, temporal demand, performance, effort, and frustration level. Additionally, a driver's pulse rate shows significant variations under different vigilance states, such as awake, drowsy, very sleepy, and fatigued (Shi et al., 2017; Geng et al., 2019). Previous studies have also utilized pulse rate as an indicator to assess driver vigilance (Sun & Yu, 2014; Mukherjee & Roy, 2024).

2.3. Evaluation method

MCDM is a tool for complex decision-making processes that involve both quantitative and qualitative factors. MCDM originated between the 1950 s and 1960 s, and its development accelerated during the 1980 s and 1990 s. Over the past few decades, it has become widely used for evaluating and ranking alternatives in various fields (Zavadskas et al., 2016). Common MCDM methods include TOPSIS, fuzzy comprehensive evaluation, and the RSR method. These three methods and their related research are described as follows.

The TOPSIS method, proposed by Hwang & Yoon (1981), is capable of fully utilizing raw data and provides results that accurately reflect the differences between evaluation alternatives. The calculation process of the TOPSIS method is relatively simple. Liang & Zhao (2020) employed the Entropy-TOPSIS method to comprehensively evaluate the operational performance of rail transit in six typical cities. Noughabi et al. (2020) applied the TOPSIS method to evaluate and rank the fatigue factors among heavy vehicle drivers. Yan et al. (2014) used TOPSIS to assess the comprehensive performance of lithium-ion batteries and found that the performance results were consistent with experimental data. However, the accuracy of the TOPSIS method may be compromised by abnormal data, which can lead to unreliable results and make it difficult to account for the weight differences among the indicators.

The fuzzy comprehensive evaluation method is employed to conduct an overall evaluation of objects or phenomena constrained by multiple factors, based on the fuzzy set theory proposed by Zadeh (1965). By transforming qualitative evaluations into quantitative ones, this method effectively addresses vague and difficult-to-quantify problems. For instance, Zhu et al., (2022) evaluated coal burst incidents using both the Analytic Hierarchy Process (AHP) and fuzzy comprehensive evaluation. However, the fuzzy comprehensive evaluation models described above rely solely on expert experience to assign indicator weights, which makes them highly susceptible to human biases. To address this limitation, objective weighting methods can be incorporated into the fuzzy comprehensive evaluation model. He et al. (2021) proposed a comprehensive fuzzy evaluation model for driving behavior safety, which combines the Entropy Weight Method (EWM) and AHP to determine the weights objectively, thereby overcoming the subjectivity inherent in fuzzy comprehensive evaluation. However, the calculation process of this method is relatively complex, and it fails to address the issue of information redundancy caused by the correlation between evaluation factors.

SRS is a statistical analysis method proposed by Tian (1988), which combines the advantages of classical parametric statistics with those of modern non-parametric statistics. Additionally, the SRS method is less sensitive to outliers, making it a more robust and reliable approach. The RSR method has been widely applied in multi-indicator comprehensive evaluations in areas such as healthcare

and transportation. For instance, [Wu et al. \(2019\)](#) utilized the TOPSIS, RSR, and SRS-TOPSIS fuzzy combination method to comprehensively evaluate the medical service capacity of Chinese traditional medicine hospitals from 2008 to 2017. [Chen et al. \(2016\)](#) developed a road safety performance indicator and used the RSR method to assess road safety across EU countries. However, the application of MCDM methods to evaluate driver vigilance under varying cognitive distractions in automated driving contexts has not been explored. A limitation of the SRS method is that it may lose certain information from the original data during the calculation process, which should be considered when interpreting the results.

These three methods were selected to evaluate the vigilance levels of drivers engaged in distracted tasks from different perspectives, ensuring the rationality of the evaluation results. Each method takes into account distinct data features. TOPSIS evaluates the proximity of the evaluation object to an idealized target, thereby sorting the alternatives and accurately reflecting the differences between various evaluation schemes. The fuzzy comprehensive evaluation method transforms qualitative assessments into quantitative ones, effectively addressing fuzzy and difficult-to-quantify issues. The RSR method, through ranking transformation, generates a dimensionless statistical quantity (RSR) and sorts or classifies the evaluation objects based on their RSR values.

3. Description of data

3.1. Apparatus

The experiments were conducted on the OKTAL SYDAC CDS650 compact fix-based simulator as shown in [Fig. 1](#). The simulator recorded real-time data on vehicle dynamics and traffic environment parameters. The simulator's control computer includes a resource channel that was used to assign distraction tasks to the drivers. Cell phones were supplied for some of the distraction tasks, and DRT was used to evaluate the vigilance reaction time. The E-prime 3.0 software with a wireless reaction button was used to build the DRT. Tobii Pro Glasses 2 were worn by the drivers to collect their eye movement data, with the area of interest being the non-forward road area; and physiological equipment was used to collect blood oxygen and pulse rate data.

3.2. Participants

Thirty-one participants (15 female and 16 male) were recruited for the study. Participants' ages were between 20 and 50. All participants had valid driver's licenses, over two years of driving experience, and experience with autonomous driving experiments or using advanced driving assistance functions. A written informed consent was obtained from all participants before the experiment, and 150 RMB (about 22 USD) was paid to each participant. As driver vigilance can be influenced by factors such as medication use and medical conditions, participants were required to be in good health, free from sleep-related disorders, and have not taken any medication within one week prior to the experiment. Additionally, participants were instructed to refrain from consuming alcohol, coffee, or energy drinks for at least 24 h before the experiment.

3.3. Experimental design

This experiment employed a within-subject design with a single factor, where the independent variable was the task type (road condition observation task and distraction tasks). All participants completed both the road condition observation and the distraction tasks.

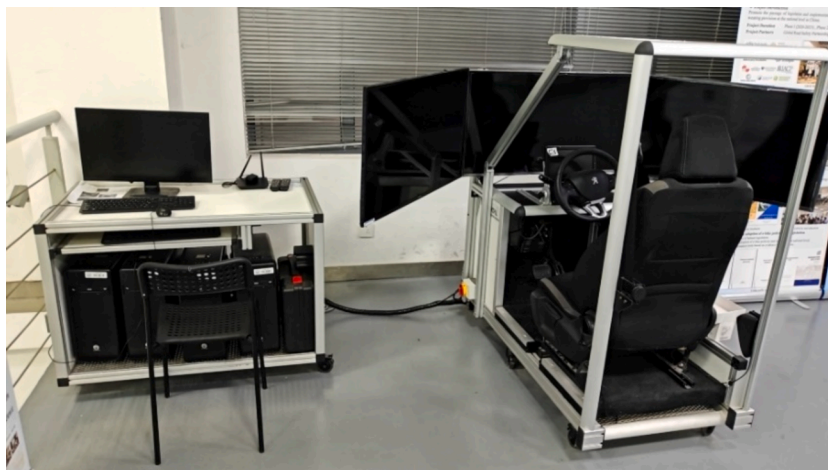


Fig. 1. Driving simulator.

3.3.1. Driving Scene setting

The driving course in the simulator experiment is a 20-km 2-way 6-lane circular highway track with a separation median as shown in Fig. 2. The parameters for the experimental road setup are in accordance with the Design Specification for highway Alignment (Ministry of Transport, 2017).

The course consisted of six straight 2-km sections (segment 1, 3, 5, 9, 11, 13), two S circular curves with radii of 700 m each (segment 7 and 15), and eight transition curves (segment 2, 4, 6, 8, 10, 12, 14 and 16). To ensure realistic simulated driving conditions and enhanced visual driving experience, a large radius S curve was used to connect the straight sections, and a gentle curve was used to connect the straight sections with the circular curve.

The configuration of the straight section is illustrated in Fig. 3. The central median has a width of 3.5 m, the lane width is 3.75 m, the paved shoulder measures 3 m in width, and the unpaved shoulder is 0.75 m wide. The total width of the roadbed is 33.5 m. Lanes are delineated by dashed lines, while solid lines are used to separate the lanes from the shoulder and the left-side curb.

The experimental vehicle and the one traveling in front of it were both autonomous vehicles, maintaining a speed of 60 km/h, which is the minimum speed limit on highways. Trees and houses are positioned along both sides of the road.

3.3.2. Road condition observation task

The primary task in which participants were engaged involved observing road conditions. Throughout the formal experiment, participants were instructed to prioritize road condition observation. Although they were not required to control the vehicle, participants were instructed to keep their left hand on the steering wheel at all times during the experiment. The road condition observation task served as the baseline (control) task for comparison with the distraction tasks.

3.3.3. Distraction tasks

This study aims to examine the impact of cognitive distraction tasks on drivers' levels of vigilance in scenarios where advanced assistance functions are used in real driving environments. In such environments, drivers are required to control the steering wheel and maintain safety monitoring of the road. To closely mimic real-life distracted driving scenarios, this study referenced a previous study on natural driving with mobile phone use, which found that the average duration of distracted driving due to mobile phone use was 69 s (Wang et al., 2020). To ensure that drivers were fully engaged in the distraction tasks, the duration of both the baseline and distraction tasks was set to 90 s. All tasks were conducted on straight road sections and were repeated twice.

In this study, distraction tasks were categorized into visual-manual-cognitive and auditory-cognitive tasks based on the driver's attention resource channels, as shown in .The auditory-cognitive tasks included the 1-back task and a news listening task. The visual-manual-cognitive tasks consisted of a typing and message-sending task, as well as a news-watching task. Following the classification of cognitive load in natural driving research (Angell et al., 2015), cognitive distraction types were further subdivided into language comprehension, language generation, working memory, and combinations of these categories.

The N-back task is a cognitive distraction task that involves working memory and decision-making (Stapel et al., 2019). It was used in this study to simulate drivers' mind-wandering, which affects driving vigilance by competing with road monitoring tasks for the driver's cognitive resources. Mind-wandering is a common phenomenon during simple tasks, such as road monitoring (Seli, 2016).

A 1-back task was used in this study as shown in Fig. 4. Drivers were asked to judge whether the current number that was broadcasted by the simulator was consistent with the previously broadcasted number. For example, when the fourth digit in the sequence is 9, the driver must determine whether the current digit (9) matches the previous digit (9) and respond with either 'yes' or 'no'. Each digit is presented at a speed of one every two seconds, and a sequence of 45 digits is randomly generated for each trial. These stimulative materials were organized in a pseudo-random order, and two distinct sets of materials (A and B) were used in the experiment. Half of the participants randomly assigned to listen to the AB sequence and the other half listening to the BA sequence.

The task of listening to news involves both auditory and cognitive distractions, with the cognitive distraction primarily related to the process of language comprehension. This type of distraction is commonly encountered in activities such as listening to radio broadcasts, audiobooks, and music. In the news listening task, participants are instructed to listen to a voice broadcast of news. A sound prompt signals the beginning and end of the task, and the material consists of two recently published news clips. Participants are randomly assigned to one of two listening orders: half listen in the AB order, while the other half listen in the BA order. The news playback was controlled by the experimenters, and the participants were not involved in any manual distractions.

Watching news task involves visual, manual and cognitive distraction, with the cognitive distraction primarily related to the process of language comprehension. Unlike the 1-back task, the cognitive process of this task relies on visual channels and is combined with manual distraction, although the manual distraction setting is relatively simple. The watching news task required participants to

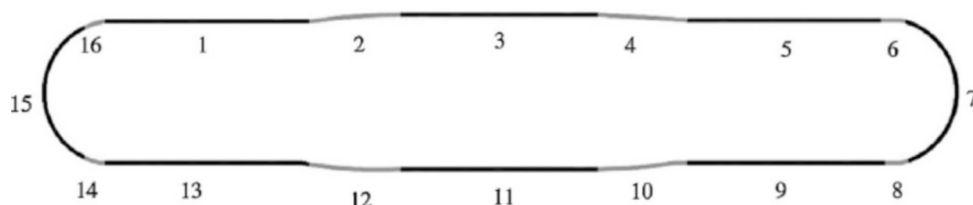


Fig. 2. Driving course with segment ID numbers.

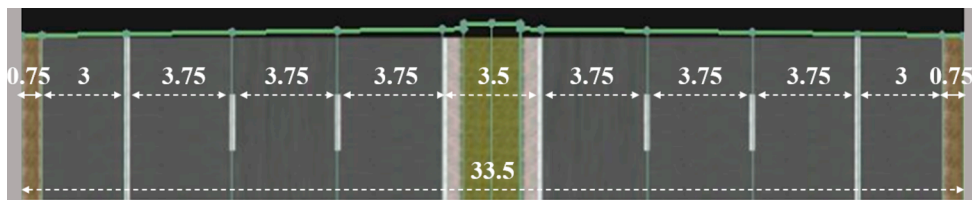


Fig. 3. Straight segments road setting.

Table 1

Task setting and analysis indicators.

Task type	Cognitive distraction type	Task	Analysis indicators
No distraction	No cognitive distraction	Baseline	1. DRT : Reaction time, Accuracy
Auditory-cognitive	Working memory	1-back	2. Eye movement : Total fixation time of road, Horizontal spread of search, Vertical spread of search
	Language-comprehension	Listening news	3. Subjective questionnaire : Mental workload
	Language-comprehension and language-generation	Watching news	4. Physiological : blood oxygen and pulse rate
Visual-manual-cognitive	Language-comprehension	Watching news	
	Language-comprehension and language-generation	Sending messages	

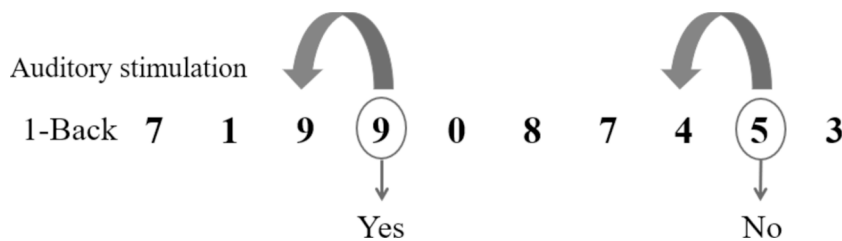


Fig. 4. 1-back task (Stapel et al., 2019).

pick up the mobile phone with their right hand. The phone was placed in the storage box in the lower right corner. Participants then needed to unlock the phone by sliding the screen. They were instructed to open WeChat and click on the news link to read. After receiving the task completion prompt, participants were told to close WeChat. Finally, they returned the phone to its original position. In a similar manner to listening to news task, this task consisted of two materials and participants were randomly assigned to the two orders (AB/BA).

Similar to the watching news task, the sending news task also involves visual, manual, and cognitive distractions. However, the

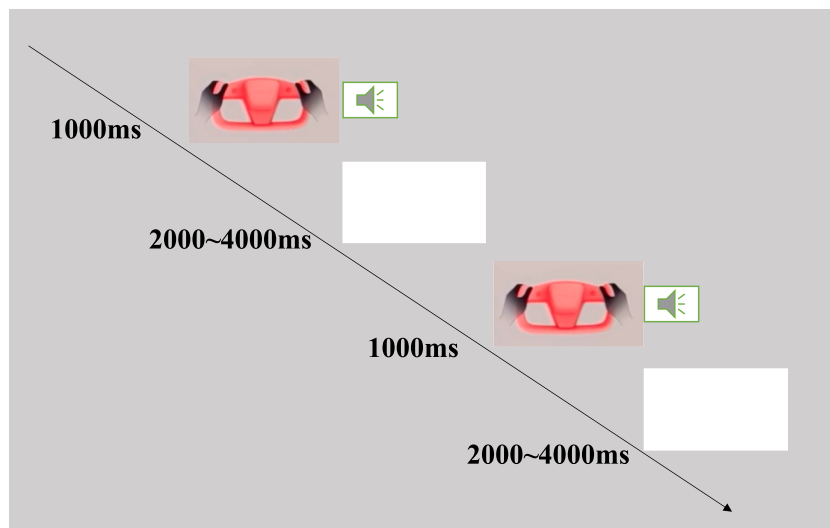


Fig. 5. DRT presentation diagram.

cognitive distractions in the sending news task are more complex, as they encompass both language comprehension and language generation processes. In sending news task, the phone was placed in the storage box in the lower right corner. Participants were instructed to pick up the supplied mobile phone with their right hand. The task then involved sliding the mobile's screen to unlock it and then to open WeChat, read a text message, and type a response to the message. The information presented to the participants consists of a simple question, such as asking where to eat next. This question is sent to the experimental mobile phone by the experimenters prior to the start of the task. When the simulator sounded a task end prompt, the drivers were instructed to send their composed messages and put the phone back in its original place. This task also consisted of two questions materials and participants were randomly assigned to the two orders (AB/BA).

3.3.4. Detection response task

The Detection Response Task (DRT) task was randomly presented every 3–5 s throughout the experiment. The DRT presentation diagram is shown in Fig. 5. The task consisted of responding to a simultaneous image-sound takeover stimulus that was randomly displayed by an external computer, which was put before the lower left side of the simulator's screen. White boxes represent that there is no information on the warning interface. Participants were instructed to respond by pressing a button (micro mouse) fixed to the steering wheel with their left thumb when they heard or saw the stimuli. They were also instructed to press the button only once upon hearing or seeing the stimulus, as continuous key pressing could render their response ineffective.

3.3.5. Experimental process

At the beginning of the experiment, participants listened to an explanation of the driving task and the purpose of the study. The total duration of experimental process which consists of six phases was approximately 90 min as shown in Fig. 6. Before starting the first phase of the formal experiment, participants were asked to complete the questionnaires on driver information and driving style, wear the physiological equipment and Tobii Pro Glasses 2 eye tracker, and practice each task.

The formal experiment was divided into two phases. Each phase presents all distraction tasks. To overcome the sequential effect, the Latin square experimental design was used to arrange six presentation orders of distraction tasks, with each 10 participants (half male and half female) using two identical sequences.

In each phase, participants made one circuit of the course for about 25 min and took a ten-minute break between phases. Participants were asked to complete three types of tasks in the formal experiment: the primary baseline task, four distraction tasks, and the DRT. They were instructed to complete a distraction task by a voice prompt, for example, "Start a watching news task." After the two formal experiments (ended), the driver filled out a subjective load scale to measure the mental workload of each distraction task and baseline.

3.4. Data collection

To analyze drivers' vigilance levels during different types of driving distractions, four types of data were collected: 1) the driver's mental workload as assessed using the NASA-TLX survey questionnaire, 2) DRT task data (reaction time and accuracy), 3) eye movement data (total fixation time on road, vertical spread of search, and horizontal spread of search), and 4) physiological data (blood oxygen and pulse rate).

The integration of data was based on the time points recorded by various devices. The driving simulation system marked the start and end time points of each task, and Python code was employed to synchronize these data.

4. Methodology

4.1. Select the indicator set for comprehensive evaluation

The random forest model was used to select the indicator set that most accurately reflects driver vigilance characteristics during various distraction tasks. This indicator set was then utilized as input for three MCDM evaluation methods. The following section outlines the process of developing these MCDM methods.

4.2. Fuzzy comprehensive evaluation model

A fuzzy comprehensive evaluation model was employed to assess the vigilance level during distracted tasks. The calculation steps are illustrated in Fig. 7. The first step of the fuzzy comprehensive evaluation model involves establishing factor sets, which are then divided using K-means clustering (evaluation set). The membership degree (affiliation matrix) is subsequently calculated. In the

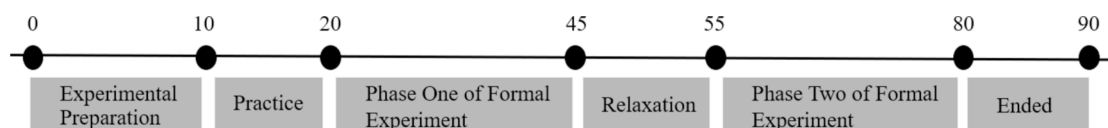


Fig. 6. Process of the experiment (min).

second step, the comprehensive weight is determined, taking into account both the subjectivity of expert evaluations and the objectivity of data features. This weight is calculated using the Analytic Hierarchy Process (AHP), the entropy weight method, and Principal Component Analysis (PCA). Finally, based on the comprehensive weight and membership degree, the comprehensive score is obtained.

4.2.1. Establish factors set

Based on the input indicators set, the factor sets were established for each evaluation indicator, in which these indicators were grouped by data type. For example,

4.2.2. Establish evaluation set

The K-means clustering algorithm was used to calculate the clustering intervals of various possible indicators to determine the range of values for each level. The evaluation standards for each indicator in the factor set A are Good, Poor, or Very Poor.

4.2.3. Affiliation matrix establishment

An affiliation matrix was constructed to assess each indicator for each distraction task. A membership degree f_n is a number between 0 and 1 that is used to indicate the degree to which the evaluation object u_0 belongs to the evaluation set A (Good, Poor, Very Poor). The term n is the number of experiments.

$$f_n = \frac{\text{Number of times A covers } u_0}{n} \quad (1)$$

4.2.4. The objective weight calculation

The objective weight w_i'' for each indicator was calculated by multiplying the factor weight η_i of the principal component analysis by the entropy weight, w_j , of each indicator as follows:

$$w_i'' = \eta_i w_j \quad (2)$$

Principal component analysis was used to establish a judgment matrix and calculate the η_i of the principal component factors. The principal components are the groups of factors, or indicators, with the most influence on the dependent variable.

The factor weight (η_i) was obtained by normalizing the variance contribution rate of each major component (η_k). Eigenvalues λ_k is the variance of the corresponding principal component. Components with λ_k greater than 1 were extracted. The larger the η_k , the greater the importance of this component. η_k for components k is calculated as follows:

$$\eta_k = \frac{\lambda_k}{\sum_{j=1}^p \lambda_k} \quad (3)$$

The entropy weight method was used to calculate the entropy weight w_j of each evaluation index, or indicator.

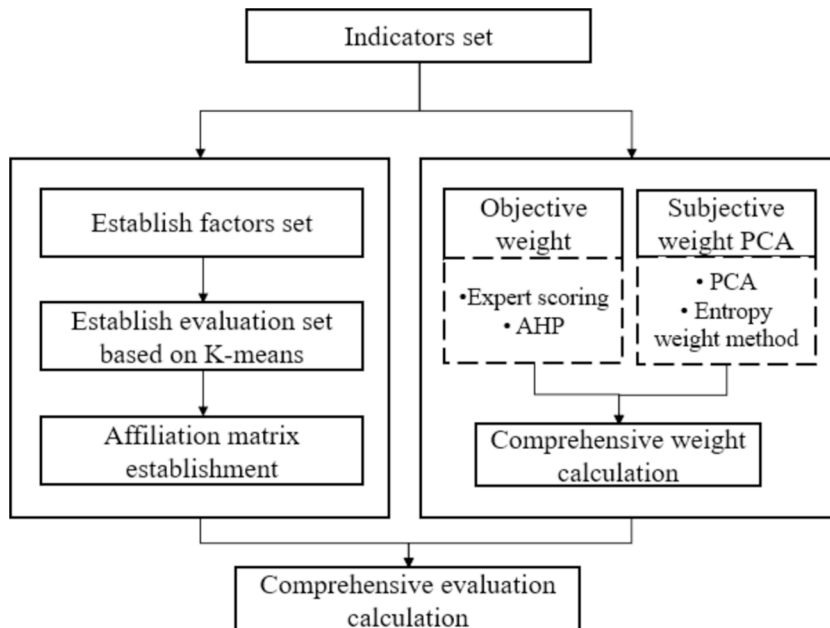


Fig. 7. The calculation steps of fuzzy comprehensive evaluation.

$$p_{ij} = x'_{ij} / \sum_{i=1}^m x'_{ij} \quad (4)$$

where m is the number of indicators, x'_{ij} is the normalized value of each indicator, and p_{ij} is the proportion of the i^{th} option of the j^{th} indicator.

e_j is the information entropy value of the j -th indicator:

$$e_j = -k \sum_{i=1}^m p_{ij} \ln(p_{ij}), (j = 1, \dots, n), k = 1/\ln(m) > 0, e_j > 0 \quad (5)$$

The entropy weight of the j -th indicator was calculated as follows:

$$w_j = 1 - e_j / \sum_{j=1}^n (1 - e_j) (j = 1, \dots, n) \quad (6)$$

4.2.5. The subjective weight calculation

The subjective weight was calculated based on the AHP. Using expert scoring of indicator weights and the summation method, subjective weight w_i' was obtained as follows:

$$w_i' = N_i / \sum_{j=1}^n N_j (i = 1, 2, \dots, m) \quad (7)$$

$$N_i = \sum_{j=1}^n \bar{x}_{ij} (i = 1, 2, \dots, m) \quad (8)$$

where x_{ij} is a judgment matrix that has been normalized for each column (each expert) of data, so that the weight of the i^{th} expert's evaluation of the j^{th} indicator is added up to 1. N_i is the vector obtained by adding x_{ij} by row (each indicator).

4.2.6. Comprehensive weight calculation

The comprehensive weight W was calculated by the weighted average of AHP weight w_i' and entropy weight w_i'' as follows:

$$W = \frac{w_i' w_i''}{\sum_{i=1}^n w_i' w_i''} \quad (9)$$

4.2.7. Comprehensive evaluation calculation

A fuzzy comprehensive evaluation model was used to derive a combined driver vigilance level for each distraction task. In Eq. (10), “ $\circ$ ” represents a fuzzy synthesis operator. T represents the degree of membership of the rated object to the fuzzy subsets of each level as a whole. Comprehensive membership degree is a number between 0 and 1, used to indicate comprehensive membership of the evaluation object (each task) to the set A (Good, Poor, Very Poor). The comprehensive evaluation T is expressed as follows:

$$T = W \circ f_n \quad (10)$$

4.3. Topsis

The basic process is based on the normalized raw data matrix, using the cosine method to identify the optimal and worst solutions among the limited options. The distance between each evaluation object and the optimal and worst solutions is then calculated separately. The relative closeness between each evaluation object and the optimal solution is derived, which serves as the basis for evaluating the quality. Assuming there are n evaluation objects (distraction tasks) and m indicators, the basic steps of the TOPSIS method are as follows:

4.3.1. Optimal and worst solutions

First, perform forward normalization and trend processing on the original data. For the positive indicators (the larger this indicator, the better the vigilance state):

$$Z_{ij} = \frac{X_{ij} - X_{\min}}{X_{\max} - X_{\min}} (i = 1, 2, 3, \dots, n; j = 1, 2, 3, \dots, m) \quad (11)$$

For the negative indicators (the smaller this indicator, the better the vigilance state):

$$Z_{ij} = \frac{X_{\max} - X_{ij}}{X_{\max} - X_{\min}} (i = 1, 2, 3, \dots, n; j = 1, 2, 3, \dots, m) \quad (12)$$

where $X_{\min}$ and $X_{\max}$ are the minimum and maximum value of the j th indicator. Then calculate the optimal and worst solutions in all

evaluation objects for each indicator:

$$\begin{aligned} \text{Optimal solutions : } (Z_1^+, Z_2^+, Z_3^+, \bullet \bullet \bullet, Z_j^+) \\ \text{Worst solutions : } (Z_1^-, Z_2^-, Z_3^-, \bullet \bullet \bullet, Z_j^-) \end{aligned} \quad (13)$$

4.3.2. Difference between each evaluation indicator and the optimal and worst Vectors

D_i^+ is the difference between each evaluation indicator and the optimal vector, D_i^- is the difference between each evaluation indicator and the worst vector. w_j is the weight of the j^{th} indicator.

$$D_i^+ = \sqrt{\sum_{j=1}^m w_j (Z_j^+ - z_{ij})^2}, D_i^- = \sqrt{\sum_{j=1}^m w_j (Z_j^- - z_{ij})^2} \quad (14)$$

4.3.3. Distance between the evaluation object and the optimal solution

C_i is the distance between the evaluation object and the optimal solution. The larger the C_i value, the better the evaluation object is (i.e., the better the vigilance state).

$$C_i = D_i^- / D_i^+ \quad (i = 1, 2, 3, \dots, n) \quad (15)$$

4.4. Rank-Sum ratio

The Rank-sum ratio (RSR) method ranks the evaluation indicators and uses the average of the ranks as the evaluation standard, which is suitable for comprehensive evaluation of indicators with different measurement units.

4.4.1. Rank matrix calculation

Assuming n evaluation objects and m evaluation indicators, construct a data matrix ($n \times m$). Improved Non integer rank sum ratio method overcomes the disadvantage of easily losing quantitative information of the original indicator value during RSR rank normalization.

For the positive indicators:

$$R_{ij} = 1 + (n - 1) \frac{X_{ij} - \min(X_{1j}, X_{2j}, \bullet \bullet \bullet, X_{nj})}{\max(X_{1j}, X_{2j}, \bullet \bullet \bullet, X_{nj}) - \min(X_{1j}, X_{2j}, \bullet \bullet \bullet, X_{nj})} \quad (16)$$

For the negative indicators:

$$R_{ij} = 1 + (n - 1) \frac{\max(X_{1j}, X_{2j}, \bullet \bullet \bullet, X_{nj}) - X_{ij}}{\max(X_{1j}, X_{2j}, \bullet \bullet \bullet, X_{nj}) - \min(X_{1j}, X_{2j}, \bullet \bullet \bullet, X_{nj})} \quad (17)$$

where R_{ij} is the rank value of the i^{th} row and j^{th} column indicators.

4.4.2. Rank sum ratio calculation

Determine the weight of each indicator based on the entropy weight method, and the calculation equation is shown in 4.1.4, formula 4. W_j is the weight of the j^{th} indicator, and the sum of weights is 1. The larger the value of RSR_i , the better the evaluation object is.

$$RSR_i = \frac{1}{n} \sum_{j=1}^m W_j R_{ij} \quad (18)$$

5. Results

5.1. Evaluation indicators

Evaluation indicators were determined using the random forest model. Eye movement indicators (total fixation time on road, vertical spread of search, and horizontal spread of search), DRT task indicators (reaction time and accuracy), subjective indicators (mental workload), and physiological indicators (blood oxygen and pulse rate) were included in the random forest model as independent variables. The dependent variable is the type of distraction task.

Table 2 presents the importance ratio of each feature. Physiological indicators, such as blood oxygen (6.70 %) and pulse rate (8.40 %), which had lower importance, were excluded from the analysis. The remaining six indicators account for more than 84.7 % of the total importance, and analyzing these six indicators is sufficient to validate the rationality of our evaluation model. By removing low-importance features, the model is simplified, thereby enhancing its interpretability and reducing computational costs. Since low-importance indicators contribute minimally to the model, their removal has only a marginal impact on model performance.

5.2. Fuzzy comprehensive evaluation model

5.2.1. Evaluation set

Using the k-means method, each indicator was clustered into the following three classifications: *Good*, *Poor*, and *Very Poor*. As shown in Table 3, *Good* reflects a nearly optimal state of driver vigilance (i.e., normal non-distracted vigilance). The indicators were classified into Good, Poor, and Very Poor clusters. The proportion of each indicator in different clusters was calculated to obtain the set determination.

5.2.2. Affiliation matrix calculation

The affiliation matrix of each indicator under different tasks is shown in Fig. 8. The baseline (no distraction) and listening to news have the highest proportion of response time (8a) and accuracy (8b) at the level “Good,” while sending messages has the highest proportion at the level “Very Poor.” For horizontal spread of search (8c), the baseline has the highest proportion of “Good” while watching news and sending messages have the highest proportions of “Very Poor.” For vertical spread of search (8d), 1-back has the highest proportion of “Good” while watching news and sending messages have the highest proportions of “Very Poor”.

For road fixation duration time (8e), 1-back and listening to news have the highest proportions of “Good” while watching news and sending messages have the highest proportions of “Very Poor”. For mental workload (8f), listening to news has the highest proportion of “Good” while sending messages has the highest proportion of “Very Poor”.

5.2.3. Comprehensive weight calculation

As shown in Table 4, the comprehensive weight w_i of each evaluation indicator was determined based on the calculated subjective weight w_i' and objective weight w_i'' . Among the indicator types, the maximum w_i was the eye movement indicator (0.436) and the minimum w_i was the subjective indicator (0.193). Among the specific indicators, the maximum w_i was the accuracy indicator (0.236) and the minimum w_i was the horizontal spread of search indicator (0.086).

5.2.4. Comprehensive evaluation calculation

The driver vigilance scores while participants were engaged in each distraction task are shown in Table 5. For the no-distraction baseline, the membership degrees of “Good”, “Poor”, and “Very Poor” were 0.609, 0.319, and 0.072, respectively. The comprehensive evaluation vigilance score of the baseline was highest (8.257). For the listening to news task and the 1-back task, the membership degrees of Good were highest, the comprehensive vigilance score of these auditory-cognitive distraction tasks were 8.470, 7.650, respectively. However, for the watching news task and sending messages tasks, the membership degrees for “Very Poor” were highest, and the comprehensive vigilance scores of these visual-manual distraction tasks were 6.257, 5.111, respectively.

5.3. TOPSIS model

The driver vigilance comprehensive score while participants were engaged in each distraction task are shown in Table 6.

5.4. RSR model

Table 7 presents the comprehensive RSR score and rank value for each indicator while participants were engaged in each distraction task. For the no-distraction baseline, the comprehensive RSR score was the highest at 0.943. For the listening to news task, the comprehensive RSR score was 0.878. For the watching news and 1-back tasks, the comprehensive RSR scores were 0.579 and 0.564, respectively. The lowest comprehensive RSR score was recorded for the sending messages task, with a value of 0.469.

The comprehensive scores of the three MCDM methods were compared, as shown in Table 8. For the no-distraction baseline, the comprehensive scores were highest across all three MCDM methods. The comprehensive scores for the listening to news task were similar to those of the baseline in all three MCDM methods. Therefore, both the no-distraction baseline and the listening to news task were classified as vigilance level 1, representing the highest level of vigilance. The comprehensive scores for the 1-back and watching news tasks were similar across the three MCDM methods and fell within the middle range, leading to their classification as vigilance level 2. The sending messages task exhibited the lowest comprehensive scores across all three MCDM methods, indicating a poorer overall vigilance state compared to the other tasks. Thus, it was classified as vigilance level 3.

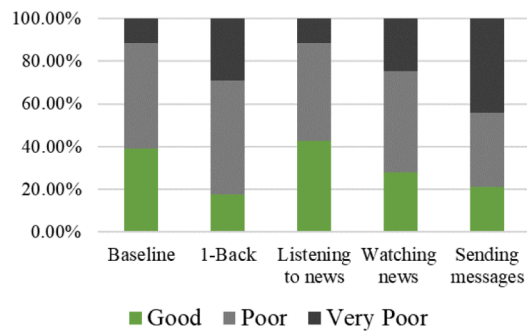
Table 2
Importance of each indicator.

Type of indicator	Specific indicator	Feature importance
Eye movement indicator	Total fixation time on road (ms)	21.50 %
	Vertical spread of search	16.90 %
	Horizontal spread of search	10.70 %
DRT task indicator	Reaction time (ms)	11.00 %
	Accuracy	10.70 %
Subjective indicator	Mental workload	13.90 %
Physiological indicator	Blood oxygen (oxi)	6.70 %
	Pulse rate (bpm)	8.40 %

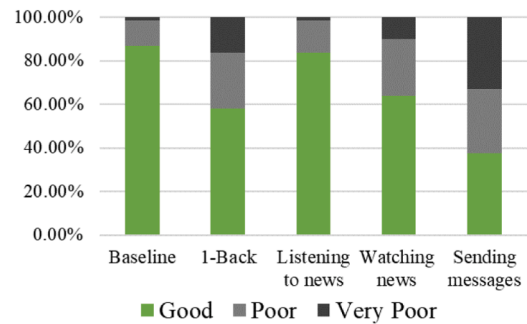
Table 3

Final clustering center based on k-means.

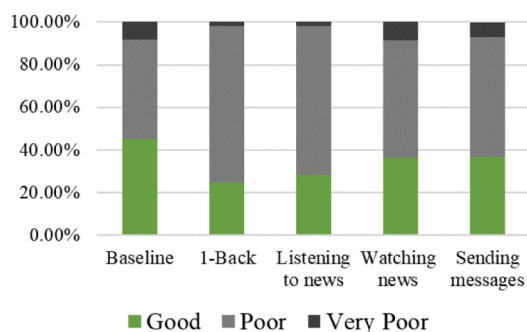
Indicators	Final clustering center			Definition of nearly optimal state of driver vigilance
	Good	Poor	Very Poor	
Total fixation time on road (ms)	30.9 %	24.8 %	44.3 %	Longer fixation time
Vertical spread of search	59.1 %	23.6 %	17.3 %	Closer to the baseline mean (116.16)
Horizontal spread of search	34.2 %	60.5 %	5.3 %	Closer to the baseline mean (63.36)
Reaction time (ms)	30.7 %	45.4 %	23.9 %	Closer to 100 ms (0 for wrong response)
Accuracy	68.8 %	19.8 %	11.4 %	Closer to 1
Mental workload	37.2 %	37.2 %	25.6 %	Closer to 0



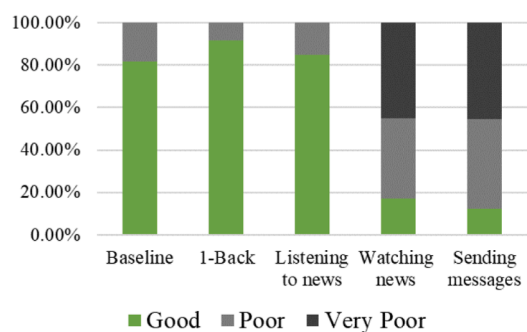
(a) Response time



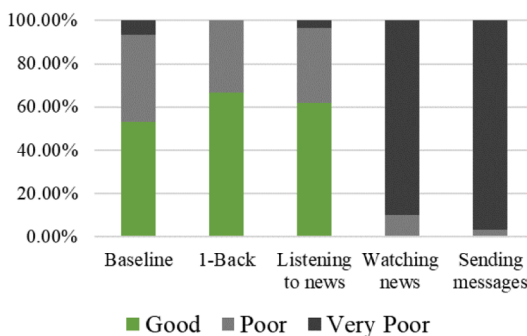
(b) Accuracy



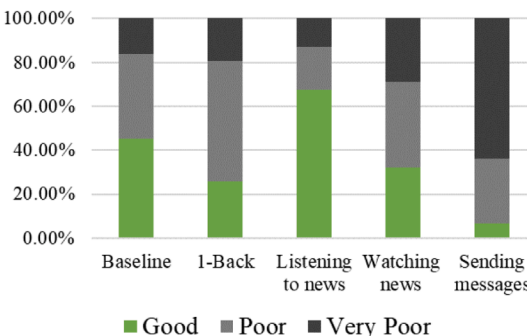
(c) Horizontal spread of search



(d) Vertical spread of search



(e) Road fixation duration time



(f) Mental workload

Fig. 8. Affiliation matrix of each task in six indicators.

Table 4
Comprehensive weights.

Type of Indicator	Specific Indicator	w_j'		w_j''		w_i	
DRT	Response time	0.375	0.194	0.306	0.107	0.371	0.136
	Accuracy		0.181		0.199		0.236
Eye movement	Road fixation duration time	0.333	0.120	0.594	0.272	0.436	0.214
	Vertical spread of search		0.102		0.204		0.136
	Horizontal spread of search		0.112		0.117		0.086
Subjective	Mental workload	0.292	0.292	0.101	0.167	0.193	0.193

The greater the weight of the indicator, the greater the its impact on vigilance.

Table 5
Fuzzy comprehensive evaluation score for each task.

Task type	Distraction task	Membership degree			Vigilance score
		Good	Poor	Very Poor	
Baseline	No distraction	0.609	0.319	0.072	8.257
Auditory-cognitive	Listening to news	0.657	0.290	0.053	8.470
	1-back	0.499	0.384	0.116	7.650
Visual-manual- cognitive	Watching news	0.305	0.321	0.373	6.257
	Sending messages	0.179	0.286	0.535	5.111

Note: Good is assigned 10 points, Poor is assigned 6 points, and Very Poor is assigned 3 points. The higher the vigilance score, the better the driver's vigilance state.

Table 6
TOPSIS evaluation method calculation results for each task.

Distraction task	Positive ideal solution distance (D +)	Negative ideal solution distance (D-)	Vigilance score index
No distraction	0.070	0.570	0.890
Listening to news	0.130	0.570	0.810
1-back	0.360	0.390	0.520
Watching news	0.360	0.380	0.510
Sending messages	0.540	0.240	0.310

Note: The higher the Comprehensive score, the better the driver's vigilance state.

Table 7
RSR evaluation method calculation results for each task and indictor.

Distraction task	Road fixation duration time	Accuracy	Mental workload	Response time	Horizontal spread of search	Vertical spread of search	Vigilance score
No distraction	4.521	4.972	4.219	4.725	0.977	4.909	0.943
Listening to news	4.830	4.999	5.000	5.000	0.764	4.056	0.878
Watching news	1.457	3.730	3.228	1.121	1.000	5.000	0.579
1-back	5.000	3.143	3.169	1.000	0.887	4.548	0.564
Sending messages	1.000	1.001	1.000	3.873	0.000	1.000	0.469

Note: The higher the RSR score, the better the driver's vigilance.

6. Discussion

As driver distraction is a critical factor affecting road safety, the inclusion of distracted driving monitoring functions in L2 vehicles is essential to prevent the misuse of advanced driving assistance systems. However, existing regulations primarily focus on physical and visual distractions associated with the use of handheld phones, with fewer restrictions on cognitive distractions. It remains unclear which types of cognitive distractions lead to significant decreases in driver vigilance. Consequently, it is uncertain which cognitive distractions should be strictly restricted, which should trigger early warnings, or which can be allowed. Therefore, this study tested various cognitive distraction tasks using a driving simulator. The fuzzy comprehensive evaluation, TOPSIS, and RSR models were employed to assess the driver vigilance levels associated with each cognitive distraction task, based on multidimensional data that

Table 8

Comparison of comprehensive scores among three MCDM methods.

Task type	Distraction task	Fuzzy comprehensive evaluation	TOPSIS	RSR	Vigilance level
No distraction	No distraction	8.257	0.890	0.943	1
Auditory-cognitive	Listening to news	8.470	0.810	0.878	1
	1-back	7.650	0.520	0.579	2
Visual-manual- cognitive	Watching news	6.257	0.510	0.564	2
	Sending messages	5.111	0.310	0.469	3

incorporate both data characteristics and expert evaluations.

6.1. Comparison of distracted tasks in different attention channels

In terms of the attention resource channels involved in distraction, visual and manual cognitive distraction tasks (watching news and sending messages) resulted in lower comprehensive vigilance scores and vigilance levels compared to auditory cognitive distraction tasks (listening to news and 1-back). Specifically, for both the listening to news and watching news tasks, which fall under the same cognitive type (language comprehension), the vigilance scores for the watching news task, which involved both visual and manual distractions, were lower across all three MCDM models compared to the listening to news task. This decrease in vigilance during the watching news task was primarily attributed to its impact on the processing of visual information on the road. As shown in Fig. 8, compared to the listening to news task, the watching news task resulted in a higher proportion of “Very Poor” ratings for drivers’ vertical spread of search and road fixation duration.

According to the resource limitation theory, attention resources and capacity are finite (Kahneman, 1973). Therefore, compared to the listening to news task, the watching news task is more likely to compete with road safety monitoring tasks for visual attention resources. Previous studies have shown that visual resources play a critical role in road safety monitoring, as 83 % of the environmental information that drivers receive is visual in nature (Trichler, 1967). This competition for visual attention results in insufficient resources for road allocation, leading to lower vigilance levels.

6.2. Comparison of the visual-manual-cognitive distraction tasks

The results show that, compared to the watching news task, performing the sending information tasks results in a lower level of vigilance, indicating that it has a greater impact on driver vigilance. Both of these tasks fall under the category of visual-manual-cognitive distractions. However, while the watching news task only involves language comprehension, the sending information task requires both language comprehension and language generation.

Language comprehension and language generation are complex and systematic cognitive processes. Neuroimaging studies have shown that these two processes involve the collaborative activation of three specific brain regions in the left hemisphere (Papathanassiou et al., 2000). Therefore, compared to the watching news task, which involves only language comprehension, the sending messages task, which requires both language comprehension and language generation, demands higher levels of brain activation and attention resources. This increase in brain capacity may limit language support, leading to longer information transmission delays (Josse et al., 2006). Consequently, individuals performing tasks that involve more extensive language processing may experience a higher cognitive workload.

The subjective mental workload perceived in the distraction tasks in this study is also consistent with this conclusion. As shown in Fig. 8, compared with the watching news task, the sending messages task exhibited a higher proportion of responses rated as “Very Poor” in terms of mental workload. Previous studies have also observed that as the mental workload of a task increases, individual volunteer performance tends to decrease (Warm et al., 2008).

6.3. Comparison of the auditory-cognitive distraction tasks

In the auditory-cognitive tasks, cognitive distractions related to working memory have a greater impact on vigilance compared to cognitive distractions related to language-comprehension (e.g., the listening to news task). As shown in Fig. 8, the 1-back task showed a higher proportion of “Very Poor” in mental workload compared to the listening news task. Because the 1-back task involved short-term working memory and required drivers’ constant comparison and judgment, mental workload of the task was higher compared to the listening news task. The impact of 1-back tasks on vigilance is also reflected in DRT tasks, as indicated by longer DRT response time, where a higher proportion of 1-back tasks had “Very Poor” response time compared to the listening to news task.

In the monotonous process of advanced assisted driving, simple auditory-cognitive tasks can help maintain driver vigilance. Among the three MCDM models, the vigilance scores for the listening to news task were similar to those for the no-distraction condition. When drivers perform simple cognitive distraction tasks while using assisted driving functions in monotonous environments, their executive function must continuously allocate resources between driving and the distraction tasks in order to mitigate the resource competition caused by automated mind-wandering (Fortenbaugh et al., 2017). In this context, the stimulus level received by the driver should not be too low. According to Yerkes-Dodson Law, the relationship between attention level and stimulus level can be described by an inverted U-shaped curve (Christianson, 1992), where moderate levels of arousal are associated with improved task performance.

Similarly, a driving simulator experiment demonstrated that, due to increased arousal, listening to music improved certain aspects of driving performance in a monotonous car-following task (Uenal et al., 2013). A previous study also found that in simple simulated traffic environments, listening to audiobooks while driving led to faster braking responses to hazards (Nowosielski et al., 2018).

6.4. The consistency of the results from the three MCDM methods

The consistency of the results obtained through the three MCDM methods supports the following three conclusions. First, the credibility of the results has been enhanced. Given the differences in data processing and computation across the three methods, this study evaluated driver vigilance during distracted tasks from various perspectives. The consistency of these results highlights the reliability and credibility of the evaluation. Second, it has been verified that the selection of evaluation indicators is stable and reliable, and the data processing and analysis conducted during the evaluation process are scientific and objective. Finally, it is worth to note that the consistency of vigilance performance in distracted tasks is maintained across multiple dimensions or indicators of performance evaluation, allowing different evaluation methods to arrive at consistent conclusions.

6.5. Limitations and future research

This study has several limitations. The first limitation is that it focuses on evaluating drivers' vigilance levels in cognitive distraction tasks within simple and monotonous driving environments. Consequently, the conclusions primarily apply to the impact of cognitive distraction on vigilance under such monotonous conditions. However, it is important to note that other types of driving environments in the real world—such as high-density urban roads and geometrically complex mountainous roads—were not considered. In these environments, the inherent complexity of driving tasks increases the demand for driver attention. Compared to the simpler and monotonous driving conditions in this study, differences in task execution and attention allocation may arise in more complex environments, which could alter cognitive load and consequently affect vigilance levels. To gain a more comprehensive understanding of the impact of cognitive distraction on vigilance levels in real-world driving conditions, future research should replicate and expand this study to include more complex driving environments.

In addition, the participant age range in this study was limited to 20 to 50 years. This primarily reflects the vigilance levels of middle-aged and young drivers when performing cognitive distraction tasks. As a result, the applicability of the study's conclusions to elderly drivers may be limited. Elderly drivers experience a decline in physical functions, especially attention and reaction time. Their vigilance may differ from that of middle-aged and young drivers. They may also be more susceptible to various factors. To further investigate the impact of cognitive distraction on driver vigilance in different age groups, future research should replicate this study with elderly drivers. This would provide a more comprehensive theoretical foundation for the design of distraction limitation and warning mechanisms for different age groups.

7. Conclusions

This study used a driving simulator to collect multidimensional data on different cognitive distraction tasks and assessed the comprehensive vigilance level of each cognitive distraction task using three multi-criteria decision-making models (fuzzy comprehensive evaluation, TOPSIS, and RSR). The main conclusions were summarized as follows. Regardless of the degree of cognitive load, tasks involving visual-manual-cognitive distraction will significantly reduce the driver's vigilance level. Therefore, visual-manual-cognitive distractions should be limited when drivers use assistance functions. Even in simple and monotonous driving environments, auditory-cognitive distractions involving working memory can lead to a significant decrease in vigilance level. Therefore, it is recommended to warn or limit auditory-cognitive distractions involving working memory. However, in monotonous and simple driving environments, auditory-cognitive distractions involving language comprehension can be allowed to improve driving performance.

This study provides theoretical support for limiting distractions by driver vigilance systems, the creation of logical rules for distraction warnings, limiting inherent distractions in the use of advanced driving assistance function, and the institution of related laws and regulations.

CRediT authorship contribution statement

Mengjiao Wu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xuesong Wang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Chris Lee:** Writing – review & editing. **Shikun Liu:** Formal analysis, Data curation. **Jiawen Chen:** Software. **Yiran Sun:** Writing – review & editing.

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Data availability

The data that has been used is confidential.

References

- Angell, L., Perez, M., & Soccolich, S. (2015). Identification of cognitive load in naturalistic driving, national surface transportation safety center for excellence. *America*.
- Chen, F., Wu, J., Chen, X., Wang, J., & Wang, D. (2016). Benchmarking road safety performance: Identifying a meaningful reference (best-in-class). *Accident Analysis and Prevention*, 86, 76–89. <https://doi.org/10.1016/j.aap.2015.10.018>
- Christianson, S. A. (1992). Emotional stress and eyewitness memory: a critical review. *Psychological Bulletin*, 112(2), 284–309. doi: 10.1037/0033-2909.111.2.284.
- Davies, D. R., & Parasuraman, R. (1982). *The psychology of vigilance*. Academic Press.
- European Union. (2019). Regulation (Eu) 2019/2144 of the European parliament and of the council. European Union.
- Fortenbaugh, F. C., Degutism, J., & Esterman, M. (2017). Recent theoretical, neural, and clinical advances in sustained attention research. *Annals of The New York Academy of Sciences*, 1396, 70–91. <https://doi.org/10.1111/nyas.13318>
- Josse, G., Hervé, P.-Y., Crivello, F., Mazoyer, B., & Tzourio-Mazoyer, N. (2006). Hemispheric specialization for language: Brain volume matters. *Brain Research*, 1068 (1), 184–193. <https://doi.org/10.1016/j.brainres.2005.11.037>
- He, S., Wang, Y., Chen, Y., & Xu, E. (2021). Research on safety evaluation of commercial vehicle driving behavior based on data mining technology. *Journal of Sensors*, 13, Article 927348. <https://doi.org/10.1155/2021/9927348>
- Hwang, C. L., & Yoon, K. (1981). *Multiple attribute decision making: Methods and applications, a state-of-the-art survey*. Berlin Heidelberg: Springer-Verlag.
- International Organization for Standardization. (2022). Road Vehicles: transport information and control systems: detection-response task (DRT) for assessing attentional effects of cognitive load in driving (iso17488y2016). *International Organization for Standardization*.
- Geng, L., Hu, Z., & Xiao, Z. (2019). Real-time fatigue driving recognition system based on deep learning and embedded platform. *American Scientific Research Journal for Engineering, Technology, and Sciences*, 53, 164–175.
- Greenlee, E. T., Delucia, P. R., & Newton, D. C. (2018). Driver vigilance in automated vehicles: Hazard detection failures are a matter of time. *Human Factors*, 60(4), 465–476. <https://doi.org/10.1177/0018720818761711>
- Karthauss, M., Wascher, E., & Falkenstein, M. (2020). The ability of young, middle-aged and older drivers to inhibit visual and auditory distraction in a driving simulator task. *Transportation Research Part F: Traffic Psychology and Behaviour*, 68, 272–284. <https://doi.org/10.1016/j.trf.2019.11.007>
- Kristina, S., & Jaka, S. (2018). Detection-response task- uses and limitations. *Sensors*, 18(2), 594. <https://doi.org/10.3390/s18020594>
- Kountouriotis, G. K., Spyridakos, P., Carsten, O. M. J., & Merat, N. (2016). Identifying cognitive distraction using steering wheel reversal rates. *Accident Analysis and Prevention*, 96, 39–45. <https://doi.org/10.1016/j.aap.2016.07.032>
- Kahneman, D. (1973). *Attention and Effort*. Englewood Cliffs, NJ: Prentice-Hall. <https://api.semanticscholar.org/CorpusID:145592484>.
- Liang, Y., & Lee, J. D. (2010). Combining cognitive and visual distraction: Less than the sum of its parts. *Accident Analysis and Prevention*, 42(3), 881–890. <https://doi.org/10.1016/j.aap.2009.05.001>
- Liang, X., & Zhao, S. C. (2020). Comprehensive evaluation of urban rail transit operation performance and diagnosis of obstacle factors based on entropy-topsis-obstacle degree. *2020 5th International Conference on Information Science, Computer Technology and Transportation*. <https://doi.org/10.1109/isctt5159.5.2020.00110>.
- Mackworth, N. (1948). The breakdown of vigilance during prolonged visual search. *Quarterly Journal of Experimental Psychology*, 1(1), 6–21. <https://doi.org/10.1080/17470214808416738>
- Mukherjee, P., & Roy, A. H. (2024). A novel deep learning-based technique for driver drowsiness detection. *Human Factors and Ergonomics in Manufacturing & Service Industries*. <https://doi.org/10.1002/hfm.21056>
- Ministry of Transport. (2017). Design Specification for highway Alignment (JTG-D20-2017). Route Design Specification People's Communications Publishing House. China.
- National Highway Traffic Safety Administration. (2012). Visual-manual NHTSA driver distraction guidelines for in-vehicle electronic devices (NHTSA-2010-0053). National Highway Traffic Safety Administration, America.
- National Center for Statistics and Analysis. (2020). Distracted driving 2019 (DOT HS 813 111). National Highway Traffic Safety Administration, America.
- Nowosielski, R. J., Trick, L. M., & Toxopeus, R. (2018). Good distractions: Testing the effects of listening to an audiobook on driving performance in simple and complex road environments. *Accident Analysis and Prevention*, 111, 202–209. <https://doi.org/10.1016/j.aap.2017.11.033>
- Noughabi, M. G., Sadeghi, A., Moghaddam, A. M., & Qazizadeh, M. J. (2020). Fatigue risk management: Assessing and ranking the factors affecting the degree of fatigue and sleepiness of heavy-vehicle drivers using TOPSIS and statistical analyses. *Iranian Journal of Science and Technology, Transactions of. Civil Engineering*, 44(4), 1345–1357. <https://doi.org/10.1007/s40996-019-00320-9>
- Papathanassiou, D., Etard, O., Mellet, E., Zago, L., Mazoyer, B., & Tzourio-Mazoyer, N. (2000). A common language network for comprehension and production: A contribution to the definition of language epicenters with PET. *NeuroImage*, 11(4), 247–357. <https://doi.org/10.1006/nimg.2000.0546>
- Posner, M. I., Rothbart, & M. K. (2007). Research on attention networks as a model for the integration of psychological science. *Annual Review of Psychology*. <https://doi.org/10.1146/annurev.psych.58.110405.085516>.
- Shi, S. Y., Tang, W. Z., & Wang, Y. Y. (2017) A review on fatigue driving detection. *The 4th Annual International Conference on Information Technology and Applications*. 12, 01019. <https://doi.org/10.1051/itmconf/20171201019>.
- Seli, P. (2016). Mind-wandering with and without intention. *Trends in Cognitive Sciences*, 20, 605–617.
- Robbins, C., & Chapman, P. (2019). How does drivers' visual search change as a function of experience? a systematic review and meta-analysis. *Accident Analysis and Prevention*, 132, Article 105266. <https://doi.org/10.1016/j.aap.2019.105266>
- Society of Automotive Engineers. (2021). Taxonomy and definitions for terms related to driving automation systems for on-road motor vehicles (SAE J3016). Society of Automotive Engineers, America.
- Son, J., & Park, M. (2021). The effects of distraction type and difficulty on older drivers' performance and behaviour: Visual vs. cognitive. *International Journal of Automotive Technology*, 22(1), 97–108.
- Sun, Y., & Yu, X. (2014). An innovative nonintrusive driver assistance system for vital signal monitoring. *IEEE J. Biomed. Health Informant.*, 18(6), 1932–1939. <https://doi.org/10.1109/JBHI.2014.2305403>
- Stapel, J., Mullakkal-Babu, F. A., & Happee, R. (2019). Automated driving reduces perceived workload, but monitoring causes higher cognitive load than manual driving. *Transportation Research Part F: Traffic Psychology and Behaviour*, 60, 590–605. <https://doi.org/10.1016/j.trf.2018.11.006>
- State Administration for Market Regulation, National standardization administration classification of automotive driving automation: GB/T 40429-2021, China. 2022.
- Tian, F. (1988). Primary investigation of composite index and RSR. *Chin. J. Health Stat.*, 4, 234–235.
- Trichler, D. G. (1967). Are you missing the boat in training aids. *Film AV Comm.*, 1, 14–16.
- Uenal, A. B., Waard, D. D., Epstude, K., & Steg, L. (2013). Driving with music: Effects on arousal and performance. *Transportation Research Part F: Traffic Psychology and Behaviour*, 21, 52–65. <https://doi.org/10.1016/j.trf.2013.09.004>
- Virtanen, K., Mansikka, H., Kontio, H., & Harris, D. (2022). Weight watchers: NASA-TLX weights revisited. *Theoretical Issues in Ergonomics Science*, 23(6), 725–748.
- Wang, H., Ding, X., & Wang, C. (2023). Risk assessment of distracted driving behavior based on visual stability coefficient. *Journal of Advanced Transportation*, 0197–6729. <https://doi.org/10.1155/2023/2111532>
- Wang, Y., Bao, S., Du, W., Ye, Z., & Sayer, J. R. (2017). Examining drivers' eye glance patterns during distracted driving: Insights from scanning randomness and glance transition matrix. *Journal of Safety Research*, 149. <https://doi.org/10.1016/j.jsr.2017.10.006>

- Warm, J. S., Parasuraman, R., & Matthews, G. (2008). Vigilance requires hard mental work and is stressful. *Human Factors*, 50(3), 433. <https://doi.org/10.1017/cbo9780511973017.018>
- Wang, X., & Xu, C. (2016). Driver drowsiness detection based on non-intrusive metrics considering individual specifics. *Accident Analysis and Prevention*, 2016(95), 350–357. <https://doi.org/10.1016/j.aap.2015.09.002>
- Wang, X., Xu, R., Asmelash, A., Xing, Y., & Lee, C. (2020). Characteristics of driver cell phone use and their influence on driving performance: A naturalistic driving study. *Accident Analysis and Prevention*, 148, Article 105845.
- World Health Organization. Global status report on road safety 2023, 20230-881.
- Wickens, C. (1984). The structure of attentional resources. *Attention & Performance Viii*, 239–257.
- Wu, X., Huang, Z., & Shenb, S. (2019). Comprehensive evaluation of medical service capability in TCM hospitals by using the fuzzy combined method of TOPSIS and Rank Sum Ratio. *IEEE International Conference on Data Science and Business Analytics*. <https://doi.org/10.1109/icdsba48748.2019.00058>
- Yan, Z., Weige, Z., Bingxiang, S., Fangdan, Z., & Man, Z. (2014). The application of TOPSIS in the study of the comprehensive performance of lithium-ion power battery. *Transportation Electrification Asia-pacific. IEEE*.
- Zadeh, L. A. (1965). *Information and control Fuzzy Sets*, 8(3), 338–353.
- Zavadskas, E. K., Mardani, A., Turskis, Z., Jusoh, A., & Nor, K. (2016). Development of Topsis method to solve complicated decision-making problems — an overview on developments from 2000 to 2015. *International Journal of Information Technology & Decision Making*, 15(03), 645–682.
- Zhang, Z. Study on the Influence of Driving Fatigue on Drivers' Effective Reaction Time. *2022 10th International Conference on Traffic and Logistic Engineering*, 2022, 9902106, 89-94.
- Zhu, Z., Wu, Y., & Han, J. (2022). A prediction method of coal burst based on analytic hierarchy process and fuzzy comprehensive evaluation. *Frontiers in Earth Science*, 9, Article 834958. <https://doi.org/10.3389/feart.2021.834958>